

1 Fourier series

$$1. x(t) = \frac{1}{\sqrt{2\pi}\sigma} \sum_{n=-\infty}^{\infty} e^{-\frac{(t-u-nT)^2}{2\sigma^2}}$$

$$\begin{aligned} a_k &= \frac{1}{T} \int_T x(\tau) e^{-j\frac{2\pi k}{T}\tau} d\tau = \frac{1}{T\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{-\left(j\frac{2\pi k}{T}\tau + \frac{(\tau-u)^2}{2\sigma^2}\right)} d\tau \\ &= \frac{1}{T\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{-\frac{(\tau-(u-j\frac{2\pi k}{T}\sigma^2))^2}{2\sigma^2}} e^{-\frac{u^2 - (u-j\frac{2\pi k}{T}\sigma^2)^2}{2\sigma^2}} d\tau \\ &= \frac{1}{T} e^{j\frac{2\pi k}{T}u} e^{\frac{(j\frac{2\pi k}{T}\sigma^2)^2}{2\sigma^2}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(\tau-(u-j\frac{2\pi k}{T}\sigma^2))^2}{2\sigma^2}} d\tau \end{aligned}$$

Gaussian integral = 1

$$\therefore a_k = \frac{1}{T} e^{j\frac{2\pi k}{T}u} e^{-\frac{2\pi^2 k^2 \sigma^2}{T^2}}$$

$$2. x(t) = |\cos(2\pi f_0 t)|, \quad T = \frac{1}{2f_0}$$

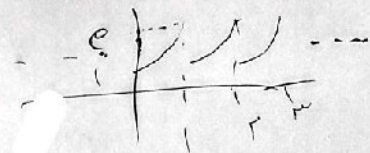
$$\begin{aligned} a_k &= 2f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} |\cos(2\pi f_0 t)| e^{-j\frac{2\pi k}{2f_0}t} dt = 2f_0 \int_0^{\frac{1}{2f_0}} \cos(2\pi f_0 t) e^{-j\pi k f_0 t} dt \\ &= 2f_0 \left[\int_0^{\frac{1}{2f_0}} e^{j\pi f_0 (1-2k)t} dt + \int_0^{\frac{1}{2f_0}} e^{-j\pi f_0 (1+2k)t} dt \right] \\ &= \left[\frac{e^{j\pi f_0 (1-2k)} - 1}{j\pi f_0 (1-2k)} - \frac{e^{-j\pi f_0 (1+2k)} - 1}{j\pi f_0 (1+2k)} \right] \end{aligned}$$

$$\rightarrow a_k = \frac{1}{\pi j} \left[\frac{e^{j\frac{\pi}{2}(1-\gamma k)} - 1}{1 - \gamma k} - \frac{e^{-j\frac{\pi}{2}(1+\gamma k)} - 1}{1 + \gamma k} \right]$$

$$= \frac{1}{\pi} \left[\frac{(-1)^k - 1}{1 - \gamma k} + \frac{(-1)^k + 1}{1 + \gamma k} \right] = \frac{\gamma}{\pi} \cdot \frac{(-1)^k}{1 - \gamma k^2}$$

$$3. x(t) = \sum_{n=-\infty}^{\infty} e^{+jn} \int u(t-n) - u(t-(n+1)) \Bigg], T=1$$

$$a_k = \int_0^1 e^{+jt} e^{-j\gamma k t} dt = \frac{e-1}{1 - \gamma j k}$$



$$4. x(t) = \cos(\gamma t) + \cos(\gamma t) = \frac{1}{\gamma} e^{j\gamma t} + \frac{1}{\gamma} e^{-j\gamma t} + \frac{1}{\gamma} e^{j\gamma t} + \frac{1}{\gamma} e^{-j\gamma t}$$

$$a_k = \begin{cases} \frac{1}{\gamma} & k = \pm 1, \pm \gamma \\ 0 & \text{o.w.} \end{cases}$$

$$5. x(t) = \sum_{n=-\infty}^{\infty} \Delta(t-n) u(t-n), \quad \Delta(t) = \text{triangle wave}$$

$$\rightarrow x(t) = \text{periodic triangle wave}$$

$$\rightarrow a_k = \int_0^1 (1-t) e^{-j\gamma k t} dt = \frac{e^{-j\gamma k} - 1}{-j\gamma k} - \left(\frac{te^{-j\gamma k}}{-j\gamma k} \Bigg|_0^1 - \int_0^1 e^{-j\gamma k t} dt \right)$$

$$= \frac{e^{-j\gamma k} - 1}{-j\gamma k} + \frac{e^{-j\gamma k}}{j\gamma k} - \frac{e^{-j\gamma k}}{-j\gamma k} = \frac{e^{-j\gamma k}}{j\gamma k}$$

$$(e^{-j\gamma k})^k = 1 \Rightarrow a_k = \frac{1}{j\gamma k}$$

2 Fourier transform | Property: ①

$$1. x(t) = \frac{t}{(a^2 + t^2)^2}$$

$$g(t) = -\frac{1}{t} \frac{1}{a^2 + t^2}$$

Property ①

$$F\left\{\frac{d}{dt} x(t)\right\} = j\omega F\{x(t)\}$$

$$\text{②: } x(t) \xrightarrow{F.T} g(\omega)$$

$$\Rightarrow g(t) \xrightarrow{F.T} \frac{1}{t\pi} x(-\omega)$$

$$F\{x\} = F\left\{\frac{d}{dt} g(t)\right\} = j\omega F\{g(t)\} \quad \star$$

$$g(t) = \left(\frac{1}{a+jt} + \frac{1}{a-jt}\right) - \frac{1}{ta}$$

$$e^{-at} u(t) \xrightarrow{F.T} \frac{1}{a-j\omega}, \quad e^{at} u(-t) \xrightarrow{F.T} \frac{1}{a+j\omega}$$

Property ②, $\frac{1}{a+jt} \xrightarrow{F.T} \left(\frac{1}{t\pi}\right)^{-1} e^{-a\omega} u(\omega), \quad \frac{1}{a-jt} \xrightarrow{F.T} \left(\frac{1}{t\pi}\right)^{-1} e^{a\omega} u(-\omega)$

$$\rightarrow F\{g(t)\} = -\frac{\pi}{ta} e^{-a|\omega|}$$

$$\star \rightarrow X(\omega) = -j\omega \frac{\pi}{ta} e^{-a|\omega|}, \quad X(t) = -\frac{j\pi}{a} e^{-a^2\pi|t|}$$

$$2. x(t) = \epsilon \operatorname{rect}\left(\frac{t}{\epsilon}\right) \cos(\pi f_0 t)$$

$$\rightarrow X(\omega) = \epsilon \times \frac{1}{\pi\epsilon} \left(F\left\{\operatorname{rect}\left(\frac{t}{\epsilon}\right)\right\} * F\{\cos(\pi f_0 t)\} \right)$$

$$= \frac{\epsilon}{\pi} \left(\epsilon \operatorname{sinc}\left(\frac{\omega}{\pi\epsilon}\right) * \pi (\delta(\omega - \omega_0) + \delta(\omega + \omega_0)) \right)$$

$$= \Lambda \left(\operatorname{sinc}\left(\epsilon \frac{\omega - \omega_0}{\pi\epsilon}\right) + \operatorname{sinc}\left(\epsilon \frac{\omega + \omega_0}{\pi\epsilon}\right) \right)$$

$$3. x(t) = \text{sinc}(t)$$

$$x(t) \xrightarrow{F.T} X(\omega) \Rightarrow \frac{1}{\omega} + x(t) \xrightarrow{F.T} -\frac{d}{d\omega} X(\omega)$$

$$x(t) \xrightarrow{F.T} X(f) \Rightarrow +x(t) \xrightarrow{F.T} \frac{j}{\pi} \frac{d}{df} X(f) \quad (1)$$

$$F\{\text{sinc}(t)\} = \text{rect}(f) \quad (2)$$

$$(1), (2), X(f) = \frac{j}{\pi} \frac{d}{df} \text{rect}(f) = \frac{j}{\pi} \left(\delta\left(f + \frac{1}{\pi}\right) - \delta\left(f - \frac{1}{\pi}\right) \right)$$

$$4. +e^{-|t|\alpha} \cos(\beta t) = x(t)$$

$$e^{-\alpha|t|} = e^{-\alpha t} u(t) + e^{\alpha t} u(-t) \xrightarrow{F.T} \frac{1}{\alpha + j\omega} + \frac{1}{\alpha - j\omega} = \frac{2\alpha}{\alpha^2 + \omega^2}$$

$$F\{e^{-|t|\alpha} \cos(\beta t)\} = \frac{1}{\pi} \left(\frac{2\alpha}{\alpha^2 + \omega^2} * \frac{1}{2} (\delta(\omega - \omega_0) + \delta(\omega + \omega_0)) \right)$$

$$= \alpha \left(\frac{1}{\alpha^2 + (\omega - \omega_0)^2} + \frac{1}{\alpha^2 + (\omega + \omega_0)^2} \right) = A(\omega)$$

$$\rightarrow X(\omega) = j \frac{d}{d\omega} A(\omega) = -\alpha j \left(\frac{\omega - \omega_0}{(\alpha^2 + (\omega - \omega_0)^2)^2} + \frac{\omega + \omega_0}{(\alpha^2 + (\omega + \omega_0)^2)^2} \right)$$

3 Type of signals

1. $x(t) = e^{-\alpha|t|} \cos(\beta t)$ even function

$$E = \lim_{T \rightarrow \infty} \int_{-T}^T |x(\tau)|^2 d\tau = \lim_{T \rightarrow \infty} \int_0^T |x(\tau)|^2 d\tau$$

$$= \lim_{T \rightarrow \infty} \int_0^T e^{-2\alpha\tau} \cos^2(\beta\tau) d\tau = \int_0^{\infty} e^{-2\alpha\tau} \left(\frac{1 + \cos(2\beta\tau)}{2} \right) d\tau$$

$$= \int_0^{\infty} e^{-2\alpha\tau} d\tau + \int_0^{\infty} e^{-2\alpha\tau} \cos(2\beta\tau) d\tau = I_1 + I_2$$

$$I_1 = \frac{e^{-2\alpha\tau}}{-2\alpha} \Big|_0^{\infty} = \frac{1}{2\alpha}, \quad I_2 = \frac{\alpha}{\alpha^2 + \beta^2}$$

$\rightarrow E = \frac{1}{2\alpha} + \frac{\alpha}{\alpha^2 + \beta^2} \rightarrow x$ is an energy signal ✓

2. $x(t) = e^{-t} \cos t$

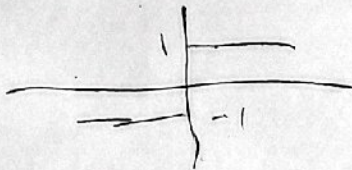
$$E = \int_{-\infty}^{\infty} e^{-2t} \cos^2 t dt = \infty$$

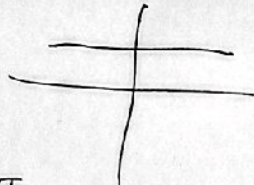
$$P = \frac{1}{T} \lim_{T \rightarrow \infty} \int_{-T}^T e^{-2t} \cos^2 t dt = \frac{1}{T} \left(\int_0^T e^{-2t} \cos^2 t dt + \int_{-T}^0 e^{-2t} \cos^2 t dt \right)$$

$$I_1 = \int_{-T}^0 \frac{1}{T} e^{-2t} dt + \int_{-T}^0 \frac{1}{T} e^{-2t} \cos^2 t dt$$

$$m_1 = \frac{e^{2T} - 1}{2}$$

$\rightarrow P = \lim_{T \rightarrow \infty} \left(\frac{e^{2T}}{2T} + \dots \right) = \infty \rightarrow$ neither a power signal nor an energy signal ✓

3. $x(t) = \text{sign}(t) =$ 

$\rightarrow |x(t)|^2 =$  $= 1 \rightarrow E = \infty$

$P = \frac{1}{T} \lim_{T \rightarrow \infty} \int_{-T}^T |x(\tau)|^2 d\tau = \frac{1}{T} \lim_{T \rightarrow \infty} \int_{-T}^T 1 d\tau$

$= \frac{1}{T} \lim_{T \rightarrow \infty} \tau \Big|_{-T}^T = \lim_{T \rightarrow \infty} \frac{T}{T} = 1 \checkmark$
Power signal

4. $x(t) = \begin{cases} K + \frac{1}{t} & t > 0 \\ 0 & t \leq 0 \end{cases}$

$E = \lim_{T \rightarrow \infty} \int_{-T}^T |x(t)|^2 dt = K \int_0^{\infty} \left(\frac{1}{t}\right)^2 dt = K \left[-\frac{1}{t} \right]_0^{\infty}$
 $= \infty$

$P = \frac{1}{T} \lim_{T \rightarrow \infty} \int_0^T K + \frac{1}{t} dt = K \lim_{T \rightarrow \infty} \frac{T}{T} = K \lim_{T \rightarrow \infty} 1 = K$

$\rightarrow$ neither an energy signal nor a power signal

5. $x(t) = \frac{1}{\pi t}$

$E = \lim_{T \rightarrow \infty} \frac{1}{\pi} \int_{-T}^T \tau^{-2} d\tau = \frac{1}{\pi} \lim_{T \rightarrow \infty} \left(-\frac{1}{\tau} \right) \Big|_{-T}^T = \infty$

$P = \frac{1}{T} \lim_{T \rightarrow \infty} \int_{-T}^T \tau^{-2} d\tau = \frac{1}{\pi} \lim_{T \rightarrow \infty} \left(-\frac{1}{\tau} \right) \Big|_{-T}^T = \infty$

$\rightarrow$ neither power nor energy

4 Fourier Transform and read integrals

$$1. \mathcal{I} = \int_{-\infty}^{\infty} \text{sinc}^{\omega}(t) dt, \quad g(t) = \text{sinc}^{\omega}(t)$$

$$X(\omega) = \int_{-\infty}^{\infty} e^{-j\omega t} g(t) dt \rightarrow \mathcal{I} = X(0)$$

$$X(\omega) = \text{rect} * \text{rect} * \text{rect} * \text{rect} * \text{rect} := C_{\omega}(f)$$

$$C_{n+1} = \text{rect} * C_n = \int_{t-\omega/\omega}^{t+\omega/\omega} C_n(\tau) d\tau$$

$$C_1 = \text{rect}(t)$$

$$C_r = 1 - |f|, \quad \text{for } |f| \leq 1, \quad \text{o.w.} = 0$$

$$C_{\omega} = \int_{t-\omega/\omega}^{t+\omega/\omega} (1 - |\tau|) d\tau = \frac{\omega}{\xi} - t^2 \quad \text{for } |t| \leq |\omega/\omega|$$

$$\rightarrow C_{\omega}(0)$$

$$C_{\xi}(0) = \int_{-\omega/\omega}^{\omega/\omega} C_{\omega}(\tau) d\tau = \frac{\omega}{\omega}$$

$$C_{\omega}(0) = \int_{-\omega/\omega}^{\omega/\omega} C_{\xi}(\tau) d\tau = \int_{-\omega/\omega}^{\omega/\omega} \int_{\tau-\omega/\omega}^{\tau+\omega/\omega} C_{\omega}(u) du d\tau$$

$$= \int_{-\infty}^{\infty} C_{\omega}(u) C_r(u) du = \omega \int_0^{\infty} C_{\omega}(u) C_r(u) du$$

$$C_{\omega} = \begin{cases} \frac{\omega}{\xi} - u^2 & 0 \leq u \leq \omega/\omega \\ 1 - u + \frac{(u - \omega/\omega)^2}{\omega} & \omega/\omega < u < 1 \end{cases} \quad C_r = \begin{cases} 1 - u & 0 \leq u \leq 1 \\ 0 & u > 1 \end{cases}$$

$$\rightarrow C_{\omega}(0) = \gamma \left\{ \int_0^{\sigma/\omega} \left(\frac{\omega}{\varepsilon} - u^2 \right) (1-u) du + \int_{\sigma/\omega}^1 \left((1-u) + \frac{(u-\sigma/\omega)^2}{\gamma} \right) (1-u) du \right\}$$

$$\rightarrow C_{\omega}(0) = \gamma \left(\frac{89}{192} + \frac{14}{384} \right) = \frac{116}{384} \times \gamma = \frac{116}{192}$$

$$3. I = \int_0^{\infty} e^{-\alpha t} \sin(\pi t) dt = \int_{-\infty}^{\infty} (e^{-\alpha t} u(t)) / (\sin(\pi t)) dt$$

Parseval's theorem: $\int_{-\infty}^{\infty} x(t) \overline{y(t)} dt = \int_{-\infty}^{\infty} X(f) \overline{Y(f)} df$

$$x := e^{-\alpha t} u(t), X = \frac{1}{\alpha + j\pi f}, y := \sin(\pi t), Y = \text{rect}(f)$$

$$\rightarrow I = \int_{-\infty}^{\infty} \frac{1}{\alpha + j\pi f} \text{rect}(f) df = \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{1}{\alpha + j\pi f} df$$

$$= \frac{-j}{\pi} \ln(\alpha + j\pi f) \Big|_{-\frac{1}{2}}^{\frac{1}{2}} = -\frac{j}{\pi} \ln\left(\frac{\alpha + j\pi}{\alpha - j\pi}\right)$$

$$= -\frac{j}{\pi} (\ln|z| + j \arg(z)) = -\frac{j}{\pi} (\ln|z| + j \pi \tan^{-1}\left(\frac{\pi}{\alpha}\right))$$

$$\rightarrow I = \frac{1}{\pi} \tan^{-1}\left(\frac{\pi}{\alpha}\right)$$

$$4. I = \int_0^{\infty} e^{-\alpha t} \cos(\beta t) dt = \int_{-\infty}^{\infty} (e^{-\alpha t} u(t)) / \cos(\beta t) dt$$

Parseval's theorem, $I = \int_{-\infty}^{\infty} \left(\frac{1}{\alpha + j\pi f} \right) \left(\frac{1}{2} \left(\delta\left(f - \frac{\beta}{\pi}\right) + \delta\left(f + \frac{\beta}{\pi}\right) \right) \right) df$

$$= \frac{1}{2} \left[\frac{1}{\alpha + j\beta} + \frac{1}{\alpha - j\beta} \right] = \frac{\alpha}{\alpha^2 + \beta^2}$$

$$9. I = \int_{-\infty}^{\infty} \frac{\sin t - t \cos t}{t^3} dt$$

we put $t = \pi x \rightarrow I = \int_{-\infty}^{\infty} \frac{\sin \pi x - \pi x \cos \pi x}{\pi^3 x^3} \pi dx$
 $dt = \pi dx$

$$= \frac{1}{\pi} \int_{-\infty}^{\infty} \underbrace{\frac{\sin \pi x - \pi x \cos \pi x}{\pi x^3}}_{g(x)} dx$$

FF. $G(w) = \int_{-\infty}^{\infty} g(x) e^{-i\pi f x} dx \rightarrow I = G(f) \Big|_{f=0}$

$$g(x) = -\frac{1}{x} \frac{d}{dx} \left(\frac{\sin \pi x}{\pi x} \right) = -\frac{1}{x} \frac{d}{dx} (\text{sinc}(x))$$

$$F\{\text{sinc}(x)\} = \text{rect}(f), \quad F\left\{\frac{d}{dx} \text{sinc}(x)\right\} = j\pi f \text{rect}(f)$$

$$\rightarrow F\left\{\left(-\frac{1}{x}\right) \left(\frac{d}{dx} \text{sinc}(x)\right)\right\} = F\left\{-\frac{1}{x}\right\} * F\left\{\frac{d}{dx} \text{sinc}(x)\right\}$$

$$= (j\pi \text{sgn}(f)) * (j\pi f \text{rect}(f)) = \pi^2 \text{sgn}(f) * \text{rect}(f) f$$

$$= \pi^2 \int_{-\omega/2}^{\omega/2} x \text{sgn}(f-x) dx = \begin{cases} f^2 - \frac{1}{4} & |f| < \omega/2 \\ 0 & \text{o.w} \end{cases}$$

$$\rightarrow I = \chi(f) \Big|_{f=0} = \pi^2 x - \frac{1}{4} = -\frac{\pi^2}{4}$$

5 Energy signals

$$\left\{ x(t), y(t) \text{ are orthogonal} \Rightarrow \langle x, y \rangle = \int_{-\infty}^{\infty} x(t) y^*(t) dt = 0 \right\}$$

1. if $g(t) = x(t) + y(t)$, $\langle x, y \rangle = 0$, $\langle x, y \rangle = \langle y, x \rangle$

$$\rightarrow E_g = \lim_{T \rightarrow \infty} \int_{-T}^T (g(t))^2 dt = \int_{-\infty}^{\infty} g(t) g^*(t) dt$$

$$= \int_{-\infty}^{\infty} (x+y)(x+y)^* dt = \int_{-\infty}^{\infty} (x(t)+y(t))(x^*(t)+y^*(t)) dt$$

$$= \int_{-\infty}^{\infty} x(t) x^*(t) dt + \int_{-\infty}^{\infty} y(t) y^*(t) dt + \int_{-\infty}^{\infty} x y^* + \int_{-\infty}^{\infty} y x^*$$

$$= E_x + E_y + \underbrace{\langle x, y \rangle + \langle y, x \rangle}_0 = E_x + E_y$$

2. $g(t) = a x(t) + b y(t)$

$$\rightarrow E_g = \int_{-\infty}^{\infty} g g^* dt = \int_{-\infty}^{\infty} (a x + b y)(a^* x^* + b^* y^*) dt$$

$$= |a|^2 \int_{-\infty}^{\infty} x x^* dt + |b|^2 \int_{-\infty}^{\infty} y y^* dt + \underbrace{c_1 c_r^* \int_{-\infty}^{\infty} x y^* dt}_0 + \underbrace{c_1^* c_r \int_{-\infty}^{\infty} x^* y dt}_0$$

$$= |a|^2 E_x + |b|^2 E_y$$

$$3. \quad z(t) = x(t) \pm y(t)$$

$$E_z = \lim_{T \rightarrow \infty} \int_{-T}^T |z(t)|^2 dt = \int_{-\infty}^{\infty} z(t) z^*(t) dt$$

$$= \int_{-\infty}^{\infty} (x(t) \pm y(t)) (x^*(t) \pm y^*(t)) dt$$

$$= \int_{-\infty}^{\infty} x x^* dt + \int_{-\infty}^{\infty} y y^* dt \pm \left(\int_{-\infty}^{\infty} x y^* dt + \int_{-\infty}^{\infty} y x^* dt \right)$$

$$= E_x + E_y \pm (E_{xy} + E_{yx})$$

5 Poisson sum formula

1.

$$K = \frac{1}{T_s} \int_{T_s}^{T_s+T_s} s(t) e^{-j \frac{2\pi K}{T_s} t} dt = \frac{1}{T_s} \int_0^{T_s} \delta(t) e^{-j \frac{2\pi K}{T_s} t} dt$$

$$\frac{1}{T_s} \int_{-\infty}^{\infty} \delta(t) e^{-j \frac{2\pi K}{T_s} t} dt = \frac{1}{T_s}$$

$$s(t) = \sum_{K \in \mathbb{Z}} a_K e^{j \frac{2\pi K}{T_s} t} \rightarrow s(t) = \sum_{n \in \mathbb{Z}} \delta(t - n T_s) = \frac{1}{T_s} \sum_{n \in \mathbb{Z}} e^{j \frac{2\pi n}{T_s} t}$$

2.

$$a_k = \frac{1}{T_s} \int_{T_s} g(t) e^{-j \frac{\omega_k t}{T_s}} dt, \text{ put } g(t) = \sum_{n \in \mathbb{Z}} a(t - nT_s)$$

$$\rightarrow a_k = \frac{1}{T_s} \int_0^{T_s} \sum_{n \in \mathbb{Z}} a(t - nT_s) e^{-j \frac{\omega_k t}{T_s}} dt$$

$$= \frac{1}{T_s} \sum_{n \in \mathbb{Z}} \int_0^T a(t - nT_s) e^{-j \frac{\omega_k t}{T_s}} dt, \text{ } t - nT_s = \tau \rightarrow$$

$$= \frac{1}{T_s} \sum_{n \in \mathbb{Z}} \int_{-nT_s}^{(-n+1)T_s} a(\tau) e^{-j \frac{\omega_k}{T_s} (\tau + nT_s)} d\tau$$

$$= \frac{1}{T_s} \sum_{n \in \mathbb{Z}} \int_{nT_s}^{(n+1)T_s} a(\tau) e^{-j \frac{\omega_k}{T_s} \tau} e^{-j \omega_k n} d\tau$$

$$= \frac{1}{T_s} \int_{-\infty}^{\infty} a(\tau) e^{-j \frac{\omega_k}{T_s} \tau} d\tau = \frac{1}{T_s} X(\omega) \Big|_{\omega = \frac{\omega_k}{T_s}}$$

$$g(t) = \sum_{k \in \mathbb{Z}} a_k e^{j \frac{\omega_k t}{T_s}}$$

$$\rightarrow \sum_{n \in \mathbb{Z}} a(t - nT_s) = \sum_{k \in \mathbb{Z}} a_k e^{j \frac{\omega_k t}{T_s}} = \frac{1}{T_s} \sum_{k \in \mathbb{Z}} X\left(\frac{\omega_k}{T_s}\right) e^{j \frac{\omega_k t}{T_s}}$$

$$3. \sum_{n \in \mathbb{Z}} x(t - nT_s) = \frac{1}{T_s} \sum_{n \in \mathbb{Z}} x\left(\frac{n}{T_s}\right) e^{j 2\pi n k + \frac{t}{T_s}}$$

put $t=0$

$$\rightarrow \sum_{n \in \mathbb{Z}} x(-nT_s) = \sum_{n \in \mathbb{Z}} x(nT_s) = \frac{1}{T_s} \sum_{n \in \mathbb{Z}} x\left(\frac{n}{T_s}\right)$$

$$4. \text{ put } T_s = 1 \rightarrow \sum_{n \in \mathbb{Z}} x(n) = \sum_{n \in \mathbb{Z}} x(n)$$

$$\text{put } x(t) = e^{-\alpha|t|} \rightarrow X(f) = \frac{2\alpha}{\alpha^2 + (2\pi f)^2} \text{ (proven before)}$$

$$\begin{aligned} \rightarrow \sum_{k=-\infty}^{\infty} \frac{2\alpha}{\alpha^2 + 2\pi^2 k^2} &= \sum_{n=-\infty}^{\infty} e^{-\alpha|n|} = 1 + \sum_{n=1}^{\infty} e^{-\alpha n} + \sum_{n=-\infty}^{-1} e^{\alpha n} \\ &= 1 + \frac{2e^{-\alpha}}{1 - e^{-\alpha}} = \frac{1 + e^{-\alpha}}{1 - e^{-\alpha}} = \coth\left(\frac{\alpha}{2}\right) \end{aligned}$$

$$\rightarrow \sum_{k=-\infty}^{\infty} \frac{2\alpha}{\alpha^2 + 2\pi^2 k^2} = \coth\left(\frac{\alpha}{2}\right)$$

> Parseval Theorem

$$1. \int_0^{\infty} \frac{1}{(1+x^2)^r} dx = \frac{1}{r} \int_{-\infty}^{\infty} \frac{1}{(1+x^2)^r} dx = \frac{1}{r} I$$

even function

$$g(t) = \frac{1}{1+t^2}$$

$$\frac{1}{r} e^{-|t|} \xrightarrow{F.T} \frac{1}{1+w^2} \Rightarrow \frac{1}{1+t^2} \xrightarrow{F.T} \pi \left(\frac{1}{r} e^{-|t|} \right) = \pi e^{-|w|}$$

$$\rightarrow G(w) = \pi e^{-|w|}$$

Parseval theorem: $\int_{-\infty}^{\infty} |g(t)|^r dt = \frac{1}{r\pi} \int_{-\infty}^{\infty} |G(w)|^r dw$

$$\rightarrow I = \int_{-\infty}^{\infty} \frac{1}{(1+w^2)^r} dw = \frac{1}{r\pi} \int_{-\infty}^{\infty} e^{-r|t|} dt = \frac{1}{r\pi} J$$

$$J = \int_0^{\infty} e^{-rt} dt + \int_{-\infty}^0 e^{rt} dt = \frac{-1}{-r} + \frac{1}{r} = 1$$

$$I = \frac{\pi}{r}$$

$$\rightarrow \int_0^{\infty} \frac{1}{(1+x^2)^r} dx = \frac{\pi}{r}$$

$$2. x_1(t) = \sum_{k \in \mathbb{Z}} \alpha^{|k|} e^{j k \frac{\Omega}{r} t}$$

$$\rightarrow \omega_k = \frac{k\Omega}{r}, \quad \omega = r\Omega t \rightarrow f_k = \frac{\frac{k\Omega}{r}}{r\Omega} = \frac{k}{r}$$

$$P_{in} = \sum_{k \in \mathbb{Z}} |a_k|^2 = \sum_{k \in \mathbb{Z}} \alpha^{2|k|} = 1 + 2 \sum_{k=1}^{\infty} \alpha^{2k} = 1 + \frac{2\alpha^2}{1-\alpha^2}$$

Parseval theorem

$$\rightarrow P_{in} = \frac{1+\alpha^2}{1-\alpha^2}$$

$$P_{out} = \sum_{|k| \leq K_{max}} |a_k|^2 = 1 + 2 \sum_{k=1}^{K_{max}} \alpha^{2k} = 1 + 2\alpha^2 \frac{1-\alpha^{2K_{max}}}{1-\alpha^2}$$

$$\rightarrow P_{out} = \frac{1+\alpha^2 - 2\alpha^{2K_{max}+2}}{1-\alpha^2}$$

$$\frac{P_{out}}{P_{in}} = \frac{1+\alpha^2 - 2\alpha^{2(K_{max}+1)}}{1+\alpha^2} \geq 0.9$$

$$\rightarrow \alpha^{2(K_{max}+1)} \leq \frac{0.1}{2} (1+\alpha^2)$$

$$\rightarrow (K_{max}+1) \geq \frac{\ln(0.05(1+\alpha^2))}{\ln \alpha} \times \frac{1}{2} \times \frac{1}{\ln \alpha}$$

$$\rightarrow K_{max} \geq \frac{\ln(\sqrt{0.05(1+\alpha^2)})}{\ln \alpha} - 1$$

$$\rightarrow K_{min} = \frac{K_{max}}{\varepsilon} = \frac{1}{\varepsilon} \left(\frac{\ln(\sqrt{0.05(1+\alpha^2)})}{\ln \alpha} - 1 \right)$$